Geometric Variation Adaptive Network for Remote Sensing Image Change Detection

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***Abstract*— Change detection identifies surface changes on the Earth by comparing two images from the same area at different times. To generate smooth change maps, a common method is fusing information from neighboring areas around each pixel. While the conventional fusion methods primarily rely on fixed regular-shaped neighboring areas, which may be inadequate in capturing the diverse and irregular geometric structures of changed ground objects. To address this limitation, we propose a novel geometric variation adaptive change detector (GVA-CD), which adaptively adjusts the shape and size of neighboring areas based on the geometrical structure of ground objects. More specifically, we design a new geometric variation adaptive module (GVAM) as a component of GVA-CD. GVAM captures the structure of the ground objects to construct geometrically flexible neighboring areas for each pixel, enabling the model to adapt to different ground object structures and generate discriminative difference features. We further propose a new difference measurement module to compute the difference between the features of pre- and post-change images by leveraging the adaptive neighboring areas. In addition, the GVA-CD introduces a multistage cross-scale fusion mechanism in both feature extraction and change map generation, to enhance the scale adaption ability of the feature extraction and change map generation. Extensive experiments on three large datasets demonstrate that our GVA-CD can outperform existing methods in change detection.**

***Index Terms*— Adaptive neighboring area, change detection, difference feature, hierarchical architecture.**

I. Introduction

CHANGE detection algorithms in remote sensing tasks are used to detect changes that appear on the surface of the Earth by analyzing two images obtained at different times at the same geographical area [1], [2], [3]. An accurate change detection is critical to many applications, such as urbanization monitoring [4], [5], [6], land usage change detection [7], disaster assessment [2], [8], and environmental monitoring [9], [10], which bears great social and economic impacts.

The critical process in change detection is to generate the difference feature from bitemporal image pairs. Early works usually compute the difference feature using human-designed models, such as change vector analysis (CVA) [11], [12], principal component analysis (PCA) [13], [14], and multivariate alteration detection (MAD) [15]. However, the handcrafted features cannot cope with complex real-world scenarios due to their low robustness and insufficient semantics, which significantly limit their performance in remote sensing change detection tasks.

Recently, deep neural networks (DNNs) have been applied in change detection [16], [17], [18], [19], [20] due to their strong ability in semantic feature representation. These methods extract the feature maps using DNNs [21], [22], [23], [24], [25], then use a predefined distance metric [24], [26] to calculate the difference between the features of pre- and post-change images, and finally generate a binary change map

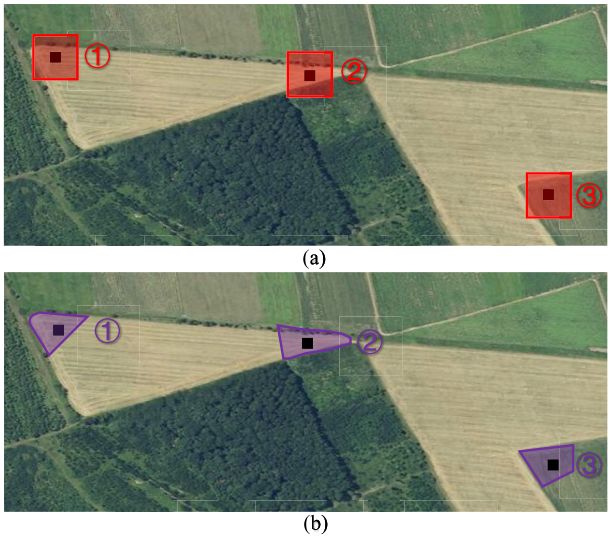

Fig. 1. Neighboring areas of the existing method and the proposed method. (a) Commonly used fixed regular-shaped neighboring areas, where the region inside the red square represents the neighboring area of the central pixel (black point). This kind of neighboring area cannot properly fit the irregular geometric shapes of ground objects in real-world remote sensing scenes. (b) Our proposed adaptive neighboring area (in purple) which can adaptively change its shape to better match the geometric shape and size of various remote sensing objects.

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based on the differences. These methods directly computed pixelwise distances between pre- and post-change feature maps to produce difference feature maps. However, these methods treated pixels as isolated entities and estimated changes based on differences between pixels, without considering that changes in real remote sensing scenes typically correspond to relatively complete objects or connected regions. As a result, the change maps produced by these methods may contain noise and lack smoothness, reflected in isolated noisy points and holes in continuous change regions.

To remedy the defect, the information from the neighboring area of a pixel should be utilized to generate a smooth difference map. A straightforward solution is to fuse the information within a regular-shaped patch centered at a pixel, as illustrated in Fig. 1(a), where the region inside the red square represents the neighboring area of the central pixel (black point). However, in real-world remote sensing scenarios, the geometric shapes of the changed ground objects are usually irregular and diverse. This diversity manifests in two aspects: 1) in various scenes, the regions where changes occur can vary significantly in their geometric characteristics, such as shape, size, and texture and 2) even in the same scene, ground objects of the same type may display differing patterns of change across different locations. The irregular and diverse geometric shapes of ground objects cannot be accurately fit by a simple regular patch. Therefore, using a fixed regular-shape neighboring area is not an optimal choice.

Recent methods employ convolutional operations to fuse neighboring information during the computation of different feature maps, ensuring a certain degree of spatial continuity and smoothness in the detected change regions. Convolution operations typically use a fixed, regular shape for the neighboring area of each pixel. However, in real-world remote sensing scenes, ground objects often have irregular and diverse geometric structures. The use of a fixed, regular neighboring area shape limits the applicability of convolution-based methods in handling such scenarios.

In this article, addressing this defect, we propose a novel change detection model that adaptively adjusts the geometry and size of neighboring areas, termed geometric variation adaptive change detector (GVA-CD). In GVA-CD, we design a geometric variation adaptive module (GVAM) that captures the geometric details of the changed regions. Specifically, it sensitively captures the geometrical structure of the ground object to search a geometrically flexible neighboring area for each pixel, as shown in Fig. 1(b). By fusing the information in the adaptive neighboring area, the proposed GVA-CD can accurately calculate difference maps, and then generate change maps that can describe the geometric properties of the changed region.

The main contributions of this article consist of threefold.

1) We propose a novel change detection model, termed GVA-CD. Unlike existing methods which mainly use fixed regular-shaped neighboring area, GVA-CD can adaptively adjust the shape and size of the neighboring areas to fit the ground objects. To achieve this goal, a new module named GVAM is designed as a component of GVA-CD. GVAM sensitively captures the geometrical structure of ground objects to construct a geometrically flexible neighboring area for each pixel. By leveraging the information in the adaptive neighboring area, GVA-CD can produce discriminative difference features and further generate accurate change maps.
2) We propose a new difference measurement module, which computes the difference by leveraging the adaptive neighboring area. This module effectively fuses the information within the neighboring area to generate accurate difference map. In addition, it is intentionally designed to be mathematically differentiable, making it suitable for integration as a component of a DNN.
3) The proposed GVA-CD introduces a new multistage cross-scale fusion mechanism to enhance the scale adaption ability of the feature extraction and change map generation. In the process of feature extraction, a pyramid-based network architecture is employed to produce scale-fused feature maps representing ground objects with varying geometrical sizes. Subsequently, during the generation of change maps, different feature maps at multiple scales are integrated to effectively capture changes of ground objects exhibiting diverse scales.

Extensive experiments on three large datasets are conducted to demonstrate the effectiveness of the proposed architecture. Compared with existing methods, the proposed GVA-CD could significantly boost the performance of capturing changes in high-resolution remote sensing images.

The remainder of this article is organized as follows. Section II discusses the related works of this study. Section III elaborates on the proposed GVA-CD. Section IV presents the experimental settings, and the experimental results on multiple remote sensing datasets. Section V concludes this article.

II. Related Work

Existing change detection methods for remote sensing images can be divided into traditional and deep learning-based methods.

A. Traditional Methods

Traditional change detection methods can be divided into pixel-based and object-based methods.

Pixel-based change detection methods generate change maps by pre- and post-change image pairs pixel by pixel. Early on, simple geometric arithmetic operations, such as image difference [27], image ratio [28], and regression analysis [29], were commonly used to obtain difference images. While these methods were straightforward, they were difficult to extract complete change information. Subsequently, in order to improve this limitation, different image generation methods based on image transformation were proposed, such as iterative weighted multivariate analysis [30], CVA [31], PCA [13], and independent component analysis (ICA) [32]. However, pixel-based methods do not consider the information within the contextual region of the pixel, causing the sensibility to noise.

The object-based change detection method employs regions extracted from images as fundamental units for change detection. It utilizes various features of the image, such as

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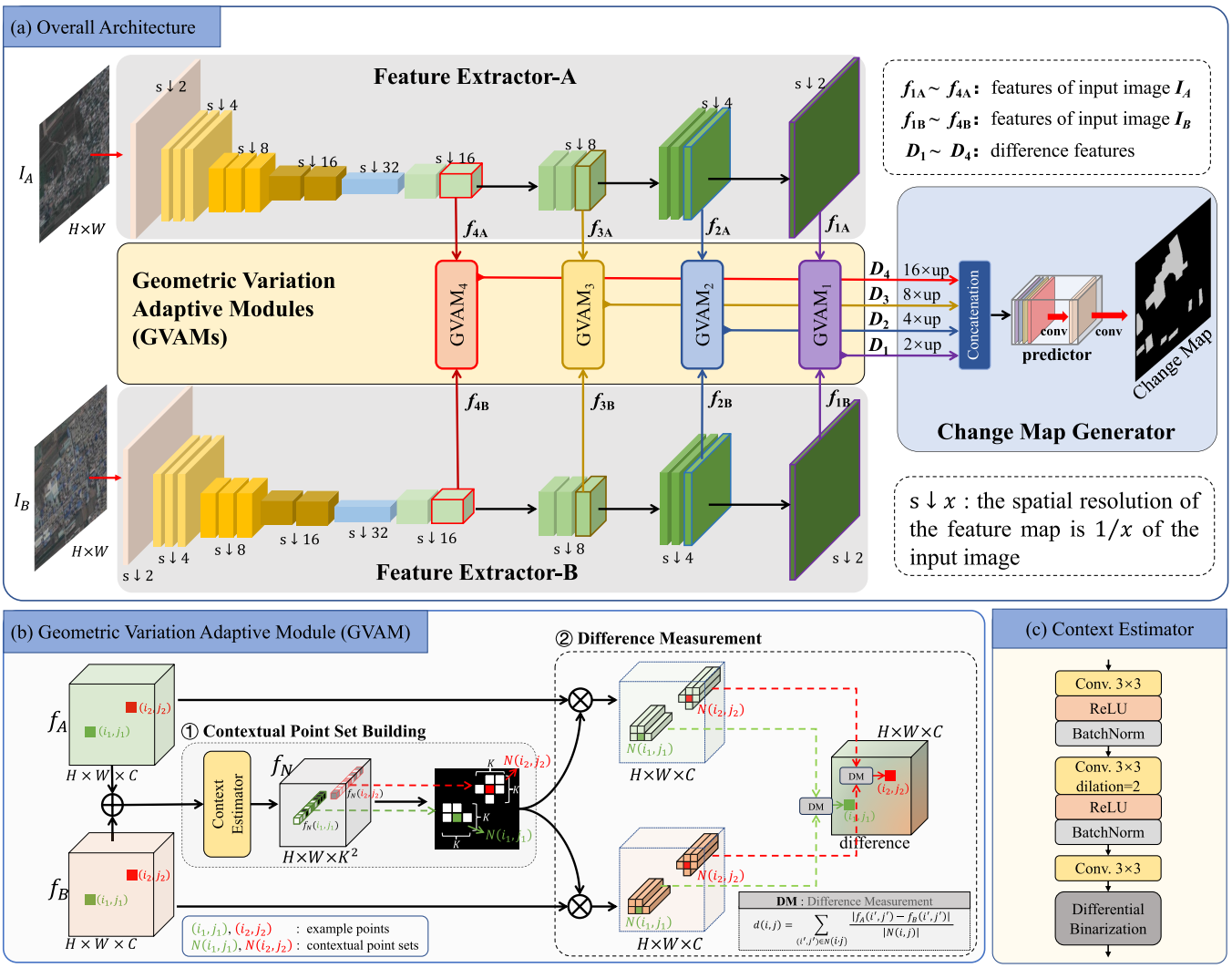

Fig. 2. Overview of the proposed GVA-CD. (a) Overall architecture of GVA-CD, which consists of: 1) a pair of weight-sharing feature extractors (Extractor-A and Extractor-B) that extract multiscale features of the pre- and post-change images (I_A and I_B), respectively; 2) GVAMs which are designed for difference measure between the features of I_A and I_B; and 3) a change map generator that is used to generate the change map. (b) Details of the proposed GVAM. (c) Details of context estimator which is an important component of GVAM.

spectral, textural, and morphological attributes, to analyze areas of change and to better model contextual information in the image. Desclée et al. [33] propose a statistical model for detecting vegetation changes based on differences in reflectance. Lefebvre et al. [34] combine the object's contour and texture from a geometric perspective for analysis to detect sudden and subtle changes. Huang et al. [35] demonstrate the robustness of brightness and texture differences between frames for detecting changes in illumination.

B. Deep Learning Based Methods

Deep learning-based change detection methods apply DNNs, e.g., convolutional neural networks (CNNs) and vision transformer (ViT), to extract high-level features of bitemporal image pairs and generate the change maps.

CNNs have shown remarkable proficiency in identifying and extracting complex patterns and features from images; therefore, they are widely used in change detection. Daudt et al. [26] developed three UNet-like [36] architectures, including FC-EF, EC-Siam-conc, and FC-Siam-diff. FC-EF utilizes early fusion strategy; that is, the bitemporal images are concatenated as a single input to the UNet-like network. EC-Siam-conc and FC-Siam-diff employ the late fusion strategy, where they first extract multiscale features separately from the input pre- and post-change images, and then fuse them using concatenation and subtraction, respectively. Zhang and Shi [22] developed a difference CNN-based change detection method, which extracts the features of bitemporal images using a pretrained VGG network [37] and computes the feature difference using a normalized L_1-distance model. Zhou et al. [38] developed a joint frequency-spatial domain network (JFSDN) to extract and integrate richer features from both spatial and frequency domains. Lei et al. [39] proposed a fully convolutional network (FCN) with a pyramidal pool, which widens a wider receptive field and fuses the multiscale features. Peng et al. [40] proposed a UNet++ network with the fusion strategy of multiple side outputs. Wang et al. [41] proposed a novel building change detection algorithm that employs cross-temporal stereo matching (CTSM) to process

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images with different viewing angles using two temporal optical spaceborne stereo images. Du et al. [42] introduced a novel concatenated deep-learning framework, named the multitask change detection network (MTCDN), for optical and SAR images, integrating a change detection network into a generative adversarial network to overcome the challenges of heterogeneity in same-ground objects. Recently, self-supervised learning framework is also introduced to change detection field; for instance, Cai et al. [43] developed a novel self-supervised method for remote sensing image change detection focusing on effective spatial feature extraction; Chen and Bruzzone [44] present a self-supervised method for pixel-level change detection in bitemporal remote sensing images and leveraging deep-learning techniques.

In recent years, ViTs have gained significant attention from change detection researchers due to their ability to capture long-range dependencies among pixels. For instance, Chen et al. [24] proposed a bitemporal image transformer that employs a transformer encoder and decoder to model the nonlocal dependencies among the CNN features extracted from the input bitemporal image pair; Bandara and Patel [25] introduced ChangeFormer, a pure-transformer architecture that directly extracts features from image patches and generates change maps using a learnable difference module; and Li et al. [23] presented TransUNetCD, which incorporates multiple transformer encoders into the UNet framework to enhance the feature representation ability of the change detection network.

III. Geometric Variation Adaptive Change Detector

In this section, we describe our proposed GVA-CD. The overall architecture of our proposed GVA-CD is shown in Fig. 2. GVA-CD mainly consists of: 1) a pair of feature extractors (Extractor-A and Extractor-B) that are utilized to extract the features of the pre- and post-change images (I_A and I_B); 2) GVAM which is designed for difference measure between the features of I_A and I_B; and 3) a change map generator that is used to generate the change map.

A. Feature Extractor

As illustrated in Fig. 3, we have designed a feature pyramid network (FPN)- [45] based feature extractor, which involves a bottom–up pathway and a top–down pathway. Through the bottom–up pathway, an ResNet-18 [46] backbone receives an input image and extracts hierarchical feature maps, denoted by $\{m_i\}_{i=1}^{5}$, where the size of m_i is $(1/2^i)$ of the input size. Then through the bottom–up pathway, the hierarchical feature maps $\{m_i\}_{i=1}^{5}$ are gradually merged by the following equation:

$$\begin{cases} f_i = \text{conv}_3\big(\text{conv}_1\big(\big[U_2(f_{i+1}); m_i\big]\big)\big), & i = 1, 2, 3, 4 \\ f_i = m_i, & i = 5 \end{cases} \tag{1}$$

where U_2 indicates $2\times$ nearest neighbor upscaling operation, $[\cdot;\cdot]$ means channel concatenation operation, and $\text{conv}_1()$, $\text{conv}_3()$ represent 1×1 and 3×3 convolution layers, respectively. Extractor-A and Extractor-B are the weight-sharing,

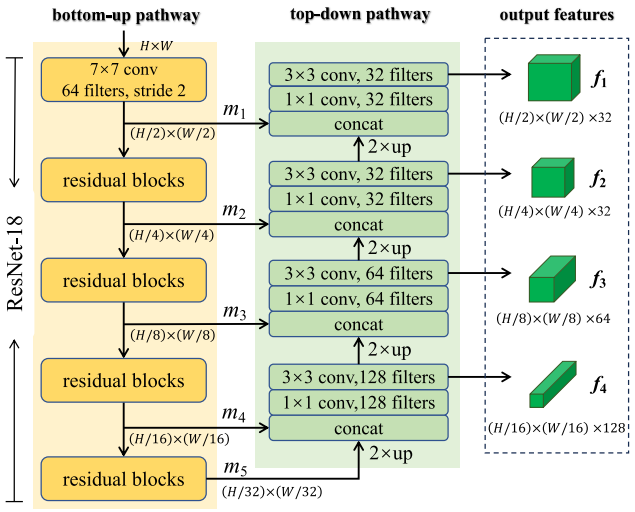

Fig. 3. Architecture of the feature extractor. The feature extractor involves a bottom–up pathway and a top–down pathway. Through the bottom–up pathway, an ResNet-18 backbone receives input images and extracts hierarchical feature maps, then through the bottom–up pathway, the hierarchical feature maps are gradually merged.

they separately receive bitemporal remote sensing images I_A and I_B, and their output features at four largest scales, denoted by $\{f_{1A}, f_{2A}, f_{3A}, f_{4A}\}$ and $\{f_{1B}, f_{2B}, f_{3B}, and f_{4B}\}$, are used to compute the change map [see Fig. 2(a)].

B. Geometric Variation Adaptive Module (GVAM) for Difference Measurement

GVAM is designed to measure the difference between the feature map of the pre- and post-change images. In our proposed GVAM, the neighboring area can adaptively change its geometric shape to fit the remote objects. Fig. 2(b) illustrates the proposed GVAM, which consists of two components: 1) a context estimator to build contextual point sets and 2) a difference measurement module.

1) Context Estimator: Prior works usually consider contextual information in a fixed regular-shaped region, which ignores the geometric structures of real-world objects. To this end, we propose a context estimator that can adaptively select the contextual region by adapting to the geometric shape of objects.

1) *Architecture of the Context Estimator:* The architecture of the context estimator is shown in Fig. 2(c). It consists of a shallow convolutional network and a differentiable binarization (DB) operator, defined as follows:

$$\text{DB}(z) = \frac{1}{1 + e^{-\eta(z-t)}} \tag{2}$$

where η indicates the amplifying factor (set to 10 empirically), and t is a learnable threshold. The DB function ensures that the output values of the context estimator are almost binarized. The mathematical property of DB is illustrated in the Appendix.

2) *Generate a Probability Tensor:* The context estimator can estimate a homogeneous region in a remote sensing image for each spatial location (i, j). Specifically, it searches the suitable points within a $K \times K$ window centered at point (i, j). The pre- and post-change

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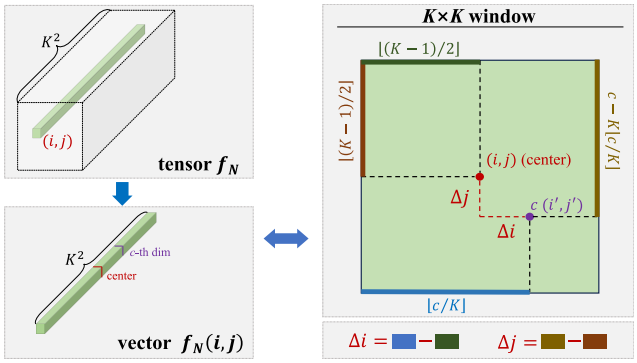

Fig. 4. Visual explanation of (3). The K^2-dim vector extracted from the spatial location (i, j) of f_N can be viewed as a $K \times K$ window centered at (i, j). The correspondence between the elements of the vector and the window are mathematically formulated by (3). The derivation of (3) is intuitively presented above.

images' feature maps are denoted by $f_A \in \mathbb{R}^{H\times W\times C}$ and $f_B \in \mathbb{R}^{H\times W\times C}$. We send the summation of f_A and f_B into the context estimator to generate a new feature tensor $f_N \in \mathbb{R}^{H\times W\times K^2}$. The tensor f_N can be considered as $H \times W$ vectors of dimension K^2, and the vector located at (i, j) is denoted by $f_N(i, j)$. Each element in $f_N(i, j)$ indicates the probability that a point in $K \times K$ window belongs to the point set of context. The $K \times K$ window is centered at (i, j) in the feature map. For example, the cth element of $f_N(i, j)$, denoted by $f_N(i, j, c)$, corresponds to the point $(i+\Delta i, j+\Delta j)$ within the $K \times K$ window, where

$$\begin{cases} \Delta i = \lfloor c/K \rfloor - \lfloor (K-1)/2 \rfloor \\ \Delta j = c - K\lfloor c/K \rfloor - \lfloor (K-1)/2 \rfloor \end{cases} \tag{3}$$

where $\lceil\cdot\rceil$ and $\lfloor\cdot\rfloor$ represent the rounding up and rounding down operations. It can be easily to deduce that, the central element on a K^2-dim vector corresponds to the elements in the $\lfloor (K-1)/2 \rfloor$ row and the $\lfloor (K-1)/2 \rfloor$ column of a $K\times K$ window. Similarly, the cth dimension element corresponds to the element in the $\lfloor c/K \rfloor$ column and $c - K\lfloor c/K \rfloor$ row of the $K \times K$ window. Therefore, it is straightforward to solve the of Δi and Δj, as shown in (3). A visual explanation of (3) is shown in Fig. 4.

3) *build Point Sets According to the Probability Tensor:* If the probability value $f_N(i, j, c)$ (almost) equals 1, the point located at $(i+\Delta i, j+\Delta j)$ is selected to build the point set of location (i, j). That is,

$$\mathcal{N}(i, j) = \{(i+\Delta i, j+\Delta j) | f_N(i, j, c) \approx 1\} \cup \{(i, j)\} \tag{4}$$

where $\mathcal{N}(i, j)$ is the built point set for location (i, j).

2) Difference Measurement: Next, we compute the difference using the above contextual point set. Given feature maps as $f_A \in \mathbb{R}^{H\times W\times C}$ and $f_B \in \mathbb{R}^{H\times W\times C}$ and denote $N(i, j)$ as the context of point (i, j), their difference $D = [d_{ijc}]^{(H,W,C)}_{(i,j,c)=(1,1,1)}$ is defined as follows:

$$d_{ijc} = \frac{\sum_{(i',j')\in\mathcal{N}(i,j)} |f_A(i', j', c) - f_B(i', j', c)|}{|\mathcal{N}(i, j)|} \tag{5}$$

where $|\mathcal{N}(i, j)|$ means the number of elements in $\mathcal{N}(i, j)$.

Equation (5) is not differentiable; thus, it cannot be directly utilized as an operator of a DNN. This is because DNNs are trained using the gradient descent method, but the gradients of a network containing a nondifferentiable function cannot be computed. For a differentiable implementation, we transform (5) in the following form:

$$\begin{aligned} d_{ijc} &= \frac{\sum_{(i',j')\in W_K(i,j)} f_{A-B}(i', j') \times \mathbb{I}\{(i, j), (i', j')\}}{\sum_{(i',j')\in W_K(i,j)} \mathbb{I}\{(i, j), (i', j')\}} \\ f_{A-B}(i', j') &= \left| f_A(i', j', c) - f_B(i', j', c) \right| \end{aligned} \tag{6}$$

where $W_K(i, j) = \{(i', j') | \max\{|i - i'|, |j - j'|\} \le K\}$ indicates a $K \times K$ window centered at (i, j), and $\mathbb{I}\{(i, j), (i', j')\}$ is a binary function whose output is 1 if $(i', j') \in \mathcal{N}(i, j)$ or 0 otherwise. As the output probability values of the context estimator are almost binarized, which is similar to the binary function $\mathbb{I}\{(i, j), (i', j')\}$, we have

$$\mathbb{I}\{(i, j), (i', j')\} \approx f_N(i, j, c) \tag{7}$$

where $c = K(i'-i+\Delta_0)+(j'-j+\Delta_0)$ and $\Delta_0 = \lfloor (K-1)/2 \rfloor$.

In our implementation, we replace the output of the binary function $\mathbb{I}\{\cdot\}$ with that of the context estimator according to (7), in order to avoid using a nondifferentiable operation.

C. Change Map Generator

1) Multiscale Difference Feature: We utilize multiple GVAM modules to compute the semantic difference of multiscale features. Specifically, we utilize a group of GVAMs, i.e., $\{\text{GVAM}_1, \text{GVAM}_2, \text{GVAM}_3, \text{GVAM}_4\}$ to measure the difference among feature pairs $\{f_{1A}, f_{1B}\}$, $\{f_{2A}, f_{2B}\}$, $\{f_{3A}, f_{3B}\}$, and $\{f_{4A}, f_{4B}\}$, respectively. That is,

$$D_i = \text{GVAM}_i(f_{iA}, f_{iB}), \quad i = 1, 2, 3, 4 \tag{8}$$

where $D_i \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times C}$ denotes the difference feature at i-th scale.

2) Predictor for Final Change Map Generation: The predictor fuses the multiscale differences to generate the final change map. It first, upscales the difference map at multiple scales to the size of the input image; then, merges these upscaled feature maps using channel concatenation operation; and finally, sends the merged feature to two successive 3×3 convolutional layers to generate the final change map.

To sum up, the predictor generates the change map by performing the following operations:

$$\begin{aligned} D_c &= [U_2(D_1); U_4(D_2); U_8(D_3); U_{16}(D_4)] \\ Y &= \text{sigmoid}\left(\text{conv}_3^1\left(\text{conv}_3^{256}(D_c)\right)\right) \end{aligned} \tag{9}$$

where Y represents the final change map, U_s $(s = 2, 4, 8, 16)$ indicates $s\times$ nearest-neighbor upscaling operation, and $\text{conv}_3^{256}(\cdot)$ and $\text{conv}_3^1(\cdot)$ denote 3×3 convolution with 256 kernels and 3×3 convolution with one kernel, respectively.

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D. Loss Function

Similar to [47], we employ a combined loss function that fuses both binary cross entropy (BCE) loss and dice loss to train our model. The BCE loss and dice loss measure the similarity between the predicted change map and the ground truth change map from different perspectives.

1) BCE Loss: BCE loss is designed for binary classification, and change detection can be considered a binary classification task of "changed" and "nonchanged" for each pixel; thus, BCE loss is used. BCE loss is defined as follows:

$$\mathcal{L}_{\mathrm{BCE}}(Y, Y^*) = -\frac{1}{N_p}\sum_{i=1}^{N_p}\left[y_i^* \log y_i + \left(1 - y_i^*\right)\log(1 - y_i)\right] \tag{10}$$

where $Y = \{y_i\}_{i=1}^{N_p}$ and $Y^* = \{y_i^*\}_{i=1}^{N_p}$ denote the predicted change map and its corresponding ground truth map, respectively; N_p means the pixel number of the map; y_i and y_i^* represent the ith pixel of the predicted change map and ground truth map; y_i^* equals 0 when the ground truth value belongs to the unchanged class, or 1 otherwise; and y_i represents the probability that the pixel is predicted as a changed class.

2) Dice Loss: The dice coefficient measures the similarity between two sets, and a change map can be regarded as a set of changed pixels. Therefore, dice loss is also applied. Dice loss is defined as follows:

$$\mathcal{L}_{\mathrm{DICE}}(Y, Y^*) = 1 - \frac{2 \times \sum_{i=1}^{N_p} y_i^* \times y_i}{\sum_{i=1}^{N_p} y_i^* + \sum_{i=1}^{N_p} y_i}. \tag{11}$$

The total loss is a weighted summation of BCE loss and dice loss, which is defined as follows:

$$\mathcal{L}(Y, Y^*) = \lambda \mathcal{L}_{\mathrm{DICE}}(Y, Y^*) + (1 - \lambda)\mathcal{L}_{\mathrm{BCE}}(Y, Y^*) \tag{12}$$

where λ is the balance weight (set to 0.5).

IV. Experimental Results and Analysis

We conduct extensive experiments to demonstrate the effectiveness of our proposed model. This section describes the experimental configurations and reports the evaluation results.

A. Experimental Configurations

1) Dataset: We carried out experiments on three public change detection datasets.

1) *LEVIR-CD Dataset:* The LEVIR-CD dataset [48] contains 445 training images, 64 validation images, and 128 test images, and each dataset contains a pair of multitemporal images with a size of 1024 × 1024. These images are from 20 different areas in several cities with a spatial resolution of 0.5 m. We cropped the image in the LEVIR-CD dataset is cropped into nonoverlapping images with the size of 256 × 256. In this way, we generate a dataset consisting of 7120 training sets, 1024 validation sets, and 2048 test sets.
2) *WHU-CD Dataset*: The WHU-CD dataset [49] consists of aerial images obtained in April 2012 and 2016. The bitemporal images contain 12 796 and 16 077 buildings in the same area, respectively. They are cropped into nonoverlapping images with the size of 256 × 256. We manually selected the effective image and generated a dataset for change detection consisting of 3167 training sets and 890 test sets.
3) *Season-Varying CD (SVCD) Dataset:* The SVCD dataset [50] contains season-varying optical satellite images of the same region from Google Earth. The 11 original remote sensing image pairs were obtained from Google Earth. The dataset consists of 16 000 clipped images with a size of 256 × 256 from original real-temporal seasonal images, distributed with 10 000 train sets, 3000 test sets also validation sets. Only image differences that correspond to the disappearance or appearance of new objects in a scene are considered in the ground-truth labels and not the differences due to seasonal factors.

2) Compared Methods: To validate the effectiveness of our proposed GVA-CD model, we compare our method with multiple deep state-of-the-art learning-based methods, including, LHBA [51], FC-EF [26], FC-Siam-conc [26], FC-Siam-diff [26], FCN-PP [39], UNet++_MOSF [40], DSIFN [52], EGRCNN [53], ICIF [54], MSPS [55], BIT-CD [24], and DGANet [56], that are described as follows.

1) *LHBA:* A traditional method, which detects the changed pixels using a local histogram-based analysis method.
2) *FC-EF:* Image-level fusion method, where the bitemporal images are concatenated as a single input to an FCN [57].
3) *FC-Siam-conc:* Feature-level fusion method, which employs a Siamese FCN to extract multilevel features and fuses bitemporal information using concatenation.
4) *FC-Siam-diff:* Feature-level fusion method, which employs a Siamese FCN to extract multilevel features and use feature difference to fuse the bitemporal information.
5) *FCN-PP:* An FCN with a pyramidal pool, which widens a wider receptive field and fuses the multiscale features.
6) *UNet++_MOSF:* A densely connected "U-shaped" convolutional network [58] with multiple side-outputs fusion.
7) *DSIFN:* An "encoder–decoder" architecture which extracts multiscale features using a VGG-based encoder and fuses feature through an attention-based decoder.
8) *EGRCNN:* An "encoder–decoder" architecture which contains two prediction branches, with one branch predicting the change area and the other predicting the edges of the change area.
9) *MSPS:* A deep multiscale Siamese network with parallel convolutional structure and self-attention [59].
10) *ICIF:* An intrascale cross-interaction and interscale feature fusion network, explicitly tapping the potential of integrating CNN and transformer [59].
11) *BIT-CD:* Transformer-based change detector, which fuses bitemporal convolutional features using transformer encoders and decoders.

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TABLE I

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT APPROACHES ON LEVIR-CD DATASET IN TERMS OF PRECISION, RECALL, F_1-SCORE, OA, AND IOU. THE BEST TWO RESULTS ARE IN **BOLD** AND UNDERLINED

method	OA	Precision	Recall	F_1-score	IoU
$LHBA_{21}$	0.8501	0.5469	0.8088	0.5307	0.3200
$FC\text{-}EF_{18}$	0.9729	0.8017	0.6173	0.6937	0.5354
$FC\text{-}Siam\text{-}conc_{18}$	0.9874	0.8860	0.8538	0.8679	0.7693
$FC\text{-}Siam\text{-}diff_{18}$	0.9862	0.9004	0.8022	0.8458	0.7374
$FCN\text{-}PP_{19}$	0.9889	0.9105	0.8514	0.8769	0.7857
$Unet++_MOSF_{19}$	0.9899	0.9149	0.8732	0.8922	0.8078
$DSIFN_{20}$	0.9875	**0.9401**	0.7895	0.8543	0.7518
$EGRCNN_{22}$	0.9892	0.8782	0.9147	0.8961	0.7890
$MSPS_{22}$	0.9903	0.9138	0.8708	0.8918	0.8136
$ICIF_{22}$	0.9899	0.9132	0.8864	0.8996	0.8175
$BIT\text{-}CD_{22}$	0.9892	0.8924	**0.8937**	0.8931	0.8068
$DGANet_{23}$	0.9903	0.9203	0.8856	0.9026	0.8225
GVA-CD (ours)	**0.9909**	0.9263	0.8788	**0.9031**	**0.8251**

The numeral in the subscript represents the publication year of the method.

TABLE II

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT APPROACHES ON WHU-CD DATASET IN TERMS OF PRECISION, RECALL, F_1-SCORE, OA, AND IOU. THE BEST TWO RESULTS ARE IN **BOLD** AND UNDERLINED

method	OA	Precision	Recall	F_1-score	IoU
$LHBA_{21}$	0.8339	0.4988	0.7914	0.5119	0.3555
$FC\text{-}EF_{18}$	0.9629	0.7456	0.7371	0.7314	0.5918
$FC\text{-}Siam\text{-}conc_{18}$	0.9752	0.8396	0.7932	0.8052	0.6959
$FC\text{-}Siam\text{-}diff_{18}$	0.9706	0.8386	0.7714	0.7934	0.6672
$FCN\text{-}PP_{19}$	0.9837	0.9200	0.8302	0.8654	0.7755
$Unet++_MOSF_{19}$	0.9839	0.8918	0.8782	0.8788	0.7968
$DSIFN_{20}$	0.9783	0.7262	0.8800	0.7921	0.6623
$EGRCNN_{22}$	**0.9903**	0.8990	0.8795	0.8890	0.7908
$MSPS_{22}$	0.9890	0.8995	0.8670	0.8829	0.7904
$ICIF_{22}$	0.9896	0.9298	0.8556	0.8832	0.7924
$BIT\text{-}CD_{22}$	0.9699	0.8850	0.9024	0.8935	0.6713
$DGANet_{23}$	0.9559	0.8659	**0.9087**	0.9026	0.8327
GVA-CD (ours)	0.9882	**0.9518**	0.8807	**0.9126**	**0.8441**

The numeral in the subscript represents the publication year of the method.

TABLE III

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT APPROACHES ON SVCD DATASET IN TERMS OF PRECISION, RECALL, F_1-SCORE, OA, AND IOU. THE BEST TWO RESULTS ARE IN **BOLD** AND UNDERLINED

method	OA	Precision	Recall	F_1-score	IoU
$FC\text{-}EF_{18}$	0.9359	0.8677	0.5435	0.6616	0.5029
$FC\text{-}Siam\text{-}conc_{18}$	0.9426	0.8524	0.6189	0.7137	0.5610
$FC\text{-}Siam\text{-}diff_{18}$	0.9427	0.8852	0.5871	0.6996	0.5470
$FCN\text{-}PP_{19}$	0.9717	0.9267	0.8024	0.8586	0.7571
$Unet++MOSF_{19}$	0.9822	0.9395	0.8973	0.9177	0.8497
$DSIFN_{20}$	0.9771	0.9496	0.8608	0.9030	0.8133
$BIT\text{-}CD_{22}$	0.9741	0.9414	**0.9224**	0.9316	0.8768
$DGANet_{23}$	0.9820	0.8659	0.9178	0.9269	0.8569
GVA-CD (ours)	**0.9859**	**0.9499**	0.9194	**0.9343**	**0.8779**

The numeral in the subscript represents the publication year of the method.

12) *DGANet:* A hybrid network architecture which incorporates two essential modules: a difference-guided aggregation module and a weighted metric module.

Among these compared methods, only LHBA, FCN-PP, and EGRCNN consider or partially consider the geometric variation of ground objects. LHBA is a traditional method utilizing a manually designed contextual model to adapt to the geometric variation of ground objects. Given that, to our knowledge, there are no deep-learning-based change detection models that fully consider geometric variation, we have selected two methods that partially consider the geometric variation, i.e., FCN-PP and EGRCNN for comparison. The

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FCN-PP method uses multiscale convolution to adapt to the scale variation of changed objects, while the EGRCNN method employs an edge-guided training strategy to encourage deep models to capture the geometric structure and contours of ground objects.

3) Evaluation Metrics: We use precision (P), recall (R), F_1-score (F_1), overall accuracy (OA), and intersection-over-union (IoU) as evaluation metrics to evaluate the performance of the proposed method. Given that TP, FP, TN, and FN represent true positives, false positives, true negatives, and false negatives, respectively. The metrics P, R, F_1, OA, and IoU are defined as follows:

$$P = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP}) \tag{13}$$

$$R = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}) \tag{14}$$

$$F_1 = (2 \times P \times R)/(P + R) \tag{15}$$

$$\mathrm{OA} = (\mathrm{TP} + \mathrm{TN})/(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}) \tag{16}$$

$$\mathrm{IoU} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP} + \mathrm{FN}). \tag{17}$$

4) Implementation Details: The training was implemented on a NVIDIA GTX 1080ti GPU with 12-GB GPU memory. We utilize Adam optimizer [60] to train the model. The learning rate is initialized to 1×10^{-4} and drops by 50% every ten epochs. The batch size is set to 8. The context window sizes (K) of $\mathrm{GVAM}_2, \mathrm{GVAM}_3, and \mathrm{GVAM}_4$ are set to 5, and that of GVAM_1 is set to 3. To reduce the overfitting effect and improve the generalization ability of the proposed network, we use random data augmentation operations in the training process, including random rotation (by $\pi/2$, π, and $3\pi/2$), and flip (in horizontal and vertical directions).

B. Quantitative Results

To evaluate the performance of the proposed GVA-CD model, we conducted experiments on the LEVIR-CD building dataset, WHU-CD dataset, and SVCD dataset. These three datasets focus on small dense ground objects, large ground objects, and season-varying changes, respectively. Thus, experiments on these three datasets can fully reflect the performance of change detection models.

1) *Results on LEVIR-CD Dataset:* To evaluate the performance of GVA-CD on intensive building changes, the comparison is made between the results achieved with GVA-CD and the several state-of-the-art methods on the LEVIR-CD dataset. Table I presents the results on the LEVIR-CD dataset. The numerical comparison results show that the proposed GVA-CD achieves the best performance with the highest OA (0.9909), F_1-score (0.9031), and IoU (0.8251). Although certain precision and recall metrics did not achieve optimal levels, it is important to note that our method still achieved superior overall performance. Precision and recall values represent specific aspects of detection performance. Precision is the proportion of true positive results among all positive predictions, indicating the accuracy of these predictions. Conversely, recall measures the proportion of actual positives correctly identified, demonstrating the method's ability to find all relevant cases. These metrics can be influenced by postprocessing thresholds and may not fully reflect the overall performance quality. The F_1-score, a harmonic mean of precision and recall, provides a more balanced evaluation of a model's performance. In our experiments, the F_1-scores across three different datasets were the highest, underlining the superior performance of our method. These results demonstrate that our GVA-CD achieves superior performance in discovering newly constructed objects. In addition, the results also show that our GVA-CD model outperforms other models that consider (or partly consider) geometric variation, such as LHBA, FCN-PP, and EGRCNN. This highlights the superiority of the GVA-CD model in handling geometric variations in change detection compared to similar methods.

2) *Results on WHU-CD Dataset:* The quantitative assessment results of the WHU-CD Dataset are presented in Table II. Our proposed GVA-CD achieves the highest scores of Precision, F_1-score, and IoU, reaching 0.9518, 0.9126, and 0.8441, respectively. Although the proposed GVA-CD model may have slightly lower scores in terms of OA and recall compared with the top-performing algorithm, it significantly outperforms all others in the remaining three metrics. Therefore, when considering the overall performance, our method emerges as the superior choice. This implies that our approach excels in accurately identifying sparse and large changes. Moreover, our GVA-CD model demonstrates superior performance in handling geometric variations in change detection, outperforming other models that consider geometric variation.

3) *Results on SVCD Dataset:* The experimental results on the SVCD dataset show the performance of our GVA-CD model in detecting the change caused by seasonal variation, we on the seasonal variation dataset. The results are shown in Table III. Our GVA-CD model outperforms all of the compared models in terms of four metrics (OA, precision, F_1-score, and IoU), which demonstrates the superiority of our method in discovering changes in natural scenes.

C. Qualitative Results

Some typical examples are selected to qualitatively compare and analyze the performance of our method and the state-of-the-art methods. The qualitative results on LEVIR-CD, WHU-CD, and SVCD datasets are shown in Figs. 5–7, respectively. For a better view, different colors are used to denote TP (white), TN (black), FP (red), and FN (green).

We can observe that our proposed GVA-CD model can consistently achieve better results compared to state-of-the-art methods. First, the GVA-CD model can better avoid the false positive [e.g., green regions in Figs. 5(I)-(III), 6(I) and (II), and 7(II) and (III)], which indicates that our model can completely detect the changed regions. As shown in Figs. 5(I) and (II) and 6(II), the compared state-of-the-art methods fail to discover the whole changed objects; whereas in Figs. 5(IV) and 7(II) and (IV), our GVA-CD can generate relatively intact prediction results when there are multiple changed regions.

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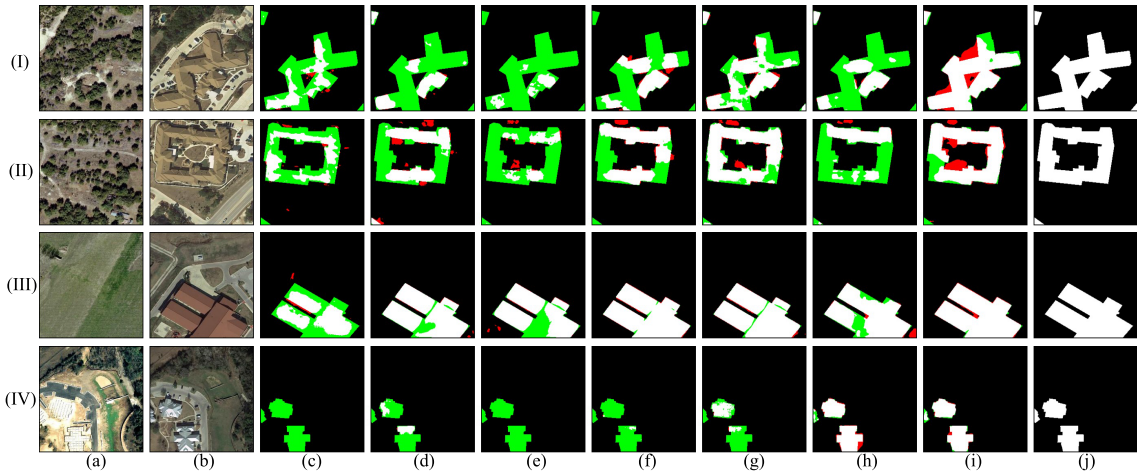

Fig. 5. Change-detection examples of different methods on the LEVIR-CD dataset. (a) Prechange image. (b) Postchange image. (c)–(i) Change maps generated by FC-EF, FC-Siam-conc, FC-Siam-diff, FCN-PP, UNet++_MOSF, BIT-CD, and our GVA-CD, respectively. (j) Ground-truth change map. Different colors are used for a better view, i.e., white for true positive, black for true negative, red for false positive, and green for false negative.

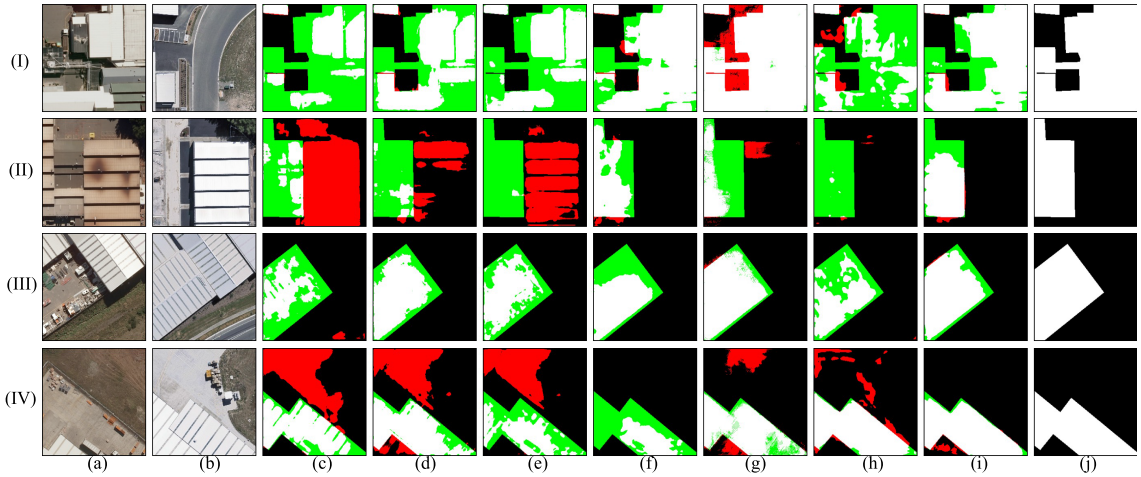

Fig. 6. Change-detection examples of different methods on the WHU-CD dataset. (a) Prechange image. (b) Postchange image. (c)–(i) Change maps generated by FC-EF, FC-Siam-conc, FC-Siam-diff, FCN-PP, UNet++_MOSF, BIT-CD, and our GVA-CD, respectively. (j) Ground-truth change map. Different colors are used for a better view, i.e., white for true positive, black for true negative, red for false positive, and green for false negative.

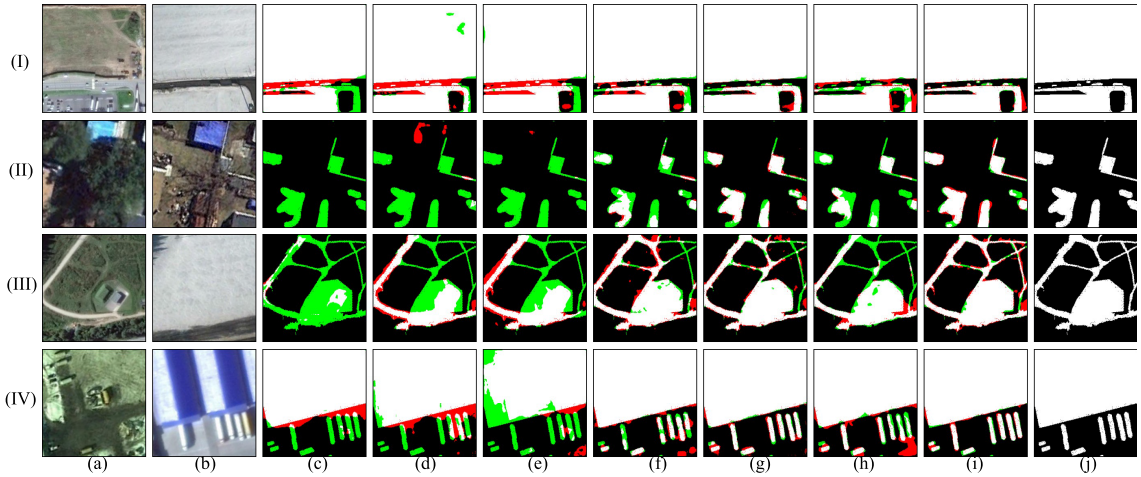

Fig. 7. Change-detection examples of different methods on the SVCD dataset. (a) Prechange image. (b) Postchange image. (c)–(i) Change maps generated by FC-EF, FC-Siam-conc, FC-Siam-diff, FCN-PP, UNet++_MOSF, BIT-CD, and our GVA-CD, respectively. (j) Ground-truth change map. Different colors are used for a better view, i.e., white for true positive, black for true negative, red for false positive, and green for false negative.

Second, the change maps detected by our method can fit the shape of the objects better than other methods [see Figs. 5(III) and 6(II)–(IV)]. The compared methods can roughly detect the changed regions, but the change maps fail to well fit the shape of the changed object in the scene: they fail to capture the detailed changes such as the boundaries in Figs. 5(III) and 6(II)–(IV), and the tiny/slender objects in Fig. 7(I) and (IV).

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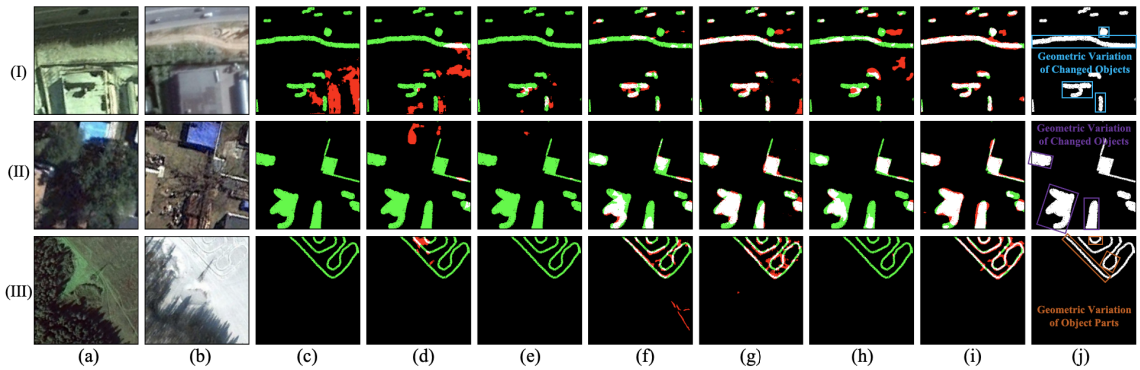

Fig. 8. Examples of change detection in remote sensing images with geometric structure variation. (a) Prechange image. (b) Postchange image. (c)–(i) Change maps generated by FC-EF, FC-Siam-conc, FC-Siam-diff, FCN-PP, UNet++_MOSF, BIT-CD, and our GVA-CD, respectively. (j) Ground-truth change map. Different colors are used for a better view, i.e., white for true positive, black for true negative, red for false positive, and green for false negative.

TABLE IV

COMPARISON OF THE CHANGE DETECTORS WITH/WITHOUT THE GVAMS ($GVAM_1$–$GVAM_4$). THE EXPERIMENTS ARE CONDUCTED ON THE LEVIR-CD DATASET. THE BEST TWO RESULTS ARE IN **BOLD** AND UNDERLINED

model	OA	Precision	Recall	F_1-score	IoU
w/o {$GVAM_1$,$GVAM_2$,$GVAM_3$,$GVAM_4$}	0.9885	0.9145	0.8470	0.8780	0.7844
w/o {$GVAM_1$,$GVAM_2$,$GVAM_3$}	0.9891	0.9165	0.8607	0.8868	0.7979
w/o {$GVAM_1$,$GVAM_2$}	0.9902	0.9009	0.8782	0.8884	0.8012
w/o {$GVAM_1$}	0.9902	0.9260	0.8707	0.8968	0.8145
GVA-CD	**0.9909**	**0.9263**	**0.8788**	**0.9031**	**0.8251**

TABLE V

COMPARISON OF CHANGE DETECTORS' PERFORMANCE USING DIFFERENCE COMPUTATIONAL MODULES WITH DIFFERENT NEIGHBORING AREAS. THE EXPERIMENTS ARE CONDUCTED ON THE LEVIR-CD DATASET. THE BEST TWO RESULTS ARE IN **BOLD** AND UNDERLINED

Neighboring Area	Difference Measure	OA	Precision	Recall	F_1-score	IoU
w/o neighboring area	L_1-distance	0.9885	0.9145	0.8470	0.8780	0.7844
fixed & regular-shaped	Conv-based	0.9891	0.9193	0.8670	0.8799	0.8179
adaptive (ours)	GVAM	**0.9909**	**0.9263**	**0.8788**	**0.9031**	**0.8251**

We have also selected typical examples that showcase geometric structure variation, as depicted in Fig. 8. Specifically, Fig. 8(I) and (II) displays this variation among multiple changed objects in a single remote scene, while Fig. 8(III) shows the geometric variation within different parts of a single changed object. The results clearly demonstrate the superiority of our GVA-CD model, which not only effectively handles these complex scenes but also yields significantly higher accuracy compared with state-of-the-art methods.

D. Ablation Study

We conduct ablation experiments to validate the effectiveness of our proposed GVAM module and adaptive neighboring area mechanism.

1) Effectiveness of GVAMs: To validate the effectiveness of the proposed GVAMs, we conducted experiments with varying numbers of GVAM and evaluated the corresponding change detection results, which are presented in Table IV. The experimental results indicate a significant decrease in the accuracy of generated change maps when our model utilizes fewer scales for difference feature calculation. Conversely, as the number of scales increases, the overall performance of the model improves. These results provide strong evidence that the design of hierarchical difference feature map generation effectively enhances the accuracy and robustness of change detection. By considering multiple scales, the design ensures accurate detection of changes in terms of both global shape and local details. This capability is essential for maintaining the reliability of our change detection model across diverse scenarios.

2) Effectiveness of Adaptive Neighboring Area: To validate the effectiveness of the proposed adaptive neighboring area mechanism, we conducted experiments by training two alternative models, in which the GVAM module was replaced with other commonly used difference feature computation modules: 1) L_1-distance based module (used by Chen et al. [24] and Daudt et al. [26]), which directly computes the L_1-distance between the feature maps of the pre- and post-change images; 2) and convolution-based module (used by Daudt et al. [26], Lei et al. [39], and Peng et al. [40]), which first concatenates the feature maps of the pre- and post-change images and then computes their difference using convolution operators. The first difference module does not consider the information in the neighboring area, while the second module considers the information in a fixed and regular shaped neighboring area.

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TABLE VI
COMPARISON OF PARAMETER NUMBER, INFERENCE TIME, AND F_1-SCORE OF DIFFERENT CHANGE DETECTION MODELS ON LEVIR-CD DATASET

Metric	FC-Siam-conc	FCN-PP	Unet++_MOSF	DSIFN	ICIF	GVA-CD
paramters amount (M)	6.26	13.4	11.0	50.4	23.82	12.97
inference time (ms)	11.8	12.5	15.3	39.4	31.4	19.8
F_1-score	0.8679	0.8769	0.8922	0.8543	0.8996	0.9031

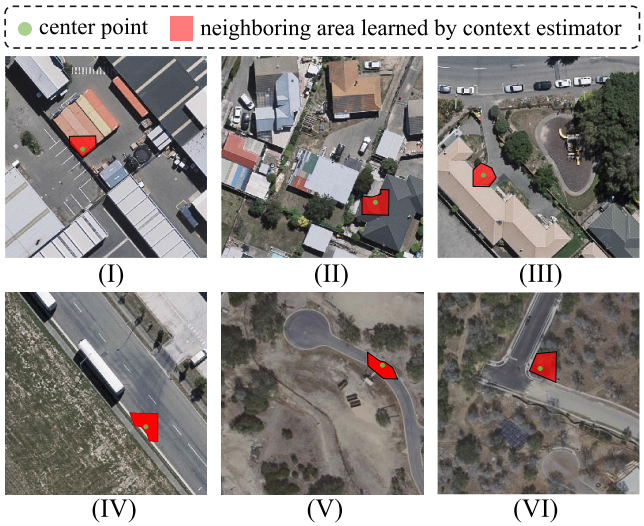

Fig. 9. Visualization of the neighboring area learned by our context estimator, our model can adaptively change the shape of the context to fit the object in remote scenarios.

The comparison results in Table V demonstrate that our proposed GVAM outperforms the aforementioned two commonly-used difference modules. This superiority can be attributed to our method's incorporation of the information in a shape-adaptive context region when calculating the differences between the features of pre- and post-change images. By considering such information, our method achieves a more precise estimation of the difference of features. In contrast, the L_1-distance based module neglects contextual information, while the convolution-based difference module restricts itself to a fixed-shaped context area. These design limitations result in reduced flexibility and adaptability, thereby contributing to their inferior performance.

E. Visualization of Learned Neighboring Area

We demonstrate the adaptability of our method to the shape of objects by visualizing the learned neighboring area using the context estimator. Fig. 9 showcases several representative examples. In Fig. 9(I)–(III), the center points correspond to buildings, and our method accurately learns neighboring areas that mostly cover the same building. In Fig. 9(IV)–(VI), the center points are located on one side of the road, and our context estimator effectively learns neighboring areas on the same side as the center points. These results highlight the adaptability of our GVAM in detecting homogeneous regions with the center point, and the learned context regions match well with the objects' shape.

F. Efficiency Analysis

To validate the efficacy of our proposed GVA-CD, we compared its parameter count and inference time with various existing methods. The inference time was measured on an NVIDIA GTX 1080ti GPU equipped with 12 GB of GPU memory, processing 256 × 256 pixel images. The comparison results are presented in Table VI. Compared to other methods, GVA-CD achieves high performance with a reasonable number of parameters and efficient inference time, striking a better balance between model size and performance.

V. CONCLUSION

A novel change detection method, called GVA-CD, has been developed to accurately detect changes in remote sensing images. Our approach overcomes the challenge of accurately depicting the geometric details of changed regions in remote scenes with large geometric variations. Specifically, our approach introduces a GVAM, which utilizes a geometrically morphable contextual model to capture the geometric structure information of the ground objects. In addition, the proposed GVA-CD is designed as a hierarchical architecture in order to enhance the robustness of our method in detecting changes at different geometric scales. Our experimental results on three large datasets demonstrate the effectiveness and generalization ability of the proposed approach. Compared with existing methods, our GVA-CD shows superior accuracy in delineating the geometric structure of changed targets, making it highly valuable for applications such as urban change detection, land-use classification, and disaster monitoring.

APPENDIX
DIFFERENTIABLE BINARIZATION FUNCTION

Fig. 10 shows the curve of the DB function defined as (2). DB behaves similar to the standard binarization, that is, when the input is greater/less than a certain threshold t, the output of DB will be very close to 1/0. When $\eta \to +\infty$, the DB function will converge with the standard binarization. That is,

$$\lim_{\eta \to +\infty} \int_{-\infty}^{+\infty} \left| \frac{1}{1+e^{-\eta(z-t)}} - B(z,t) \right| dz = 0 \tag{18}$$

where $B(z,t)$ is a standard binarization function defined as follows:

$$B(z,t) = \begin{cases} 1, & z \geq t \\ 0, & z < t \end{cases}. \tag{19}$$

The derivative of the DB function with respect to z is

$$\mathrm{DB}'(z) = \frac{\eta e^{-\eta(z-t)}}{\left(1+e^{-\eta(z-t)}\right)^2}, \quad z \in \mathbb{R}. \tag{20}$$

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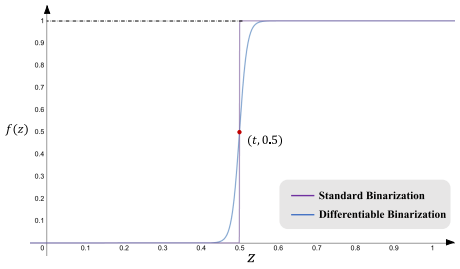

Fig. 10. Comparison of the standard binarization function and the DB function as defined by (2).

DB is differentiable throughout the entire real domain $(-\infty, +\infty)$, which enables us to directly use it as a component of DNNs.

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